# Cranberries and Heat Stress: Technical Report

STAT 628 Module 2 — Group 5

## 1 Summary

This report investigates the effects of heat stress on cranberry production using data from two field experiments conducted in western Wisconsin in 2024. The long-term experiment placed open top chambers (OTCs) over plots for the entire growing season, while the acute experiment delivered short two-hour heat pulses at approximately 40 degrees Celsius with varying frequency (1 to 4 pulses). Two cultivars were studied: Mullica Queen (MQ) and Stevens (St).

Our main findings are as follows. First, long-term OTC warming did not significantly affect either yield or gallic acid concentration; cultivar was the dominant factor in both outcomes. Second, acute heat pulses significantly reduced healthy fruit weight by 155 grams per plot (about 15%) and increased the odds of fruit rot by 3.8 times. Third, the number of heat pulses did not matter: one pulse caused the same damage as four, suggesting a threshold damage mechanism around 35 degrees Celsius. Fourth, temperature data confirmed that OTC successfully raised fruit temperature by approximately 16 degrees above control plots. Finally, gallic acid concentration was primarily driven by cultivar differences rather than heat treatment.

## 2 Introduction

Cranberries are Wisconsin's state fruit, with over half of all US cranberry production located in the state. Anecdotal evidence suggests that rising temperatures and more frequent heat events may increase fruit rot and reduce yields. To investigate this, researchers conducted two experiments examining different aspects of heat stress on cranberry production.

The **long-term experiment** placed OTC chambers over 16 plots (8 per cultivar) from early June through late September. Each OTC plot was paired with a nearby control plot that had no chamber.

The **acute experiment** delivered short, intense heat pulses using insulated OTCs with heat lamps controlled to reach 40 degrees Celsius. Four treatment levels (A through D) received 4, 3, 2, and 1 pulse(s), respectively. Each treatment had 3 biological replicates per cultivar, with paired control plots. Treatment A0 data were discarded because the heat treatment was too intense and damaged the plants.

This report addresses three questions: (1) how do heat treatments affect cranberry yield? (2) how do heat treatments affect gallic acid concentration? (3) how effectively did the experimental setup control temperature, and can actual temperature measurements help explain yield variation?

Section 3 presents the yield analysis. Section 4 covers gallic acid concentration estimation and analysis. Section 5 addresses temperature and relative humidity data. Section 6 provides conclusions.

## 3 Experimental Design and Data

### 3.1 Yield Data

The long-term yield dataset (`LTYielddata2024.csv`) contains one row per plot (n = 16) with counts and weights of healthy and rotten berries harvested from a two-square-foot area.

The acute yield dataset (`Acute HS-Yield_RawData 2024.xlsx`) contains one row per plot (n = 48, after excluding A0 treatment). Variables include cultivar, treatment (A through D or corresponding control), biological replicate number, and counts and weights of healthy and rotten fruit.

### 3.2 Phenolics Data

The file `Phenolics_RawData.xlsx` contains absorbance measurements at 765 nm. Each sample was measured three times (three lab preparation replicates). Standard samples with known concentrations (0 to 750 units) were included on each lab date to construct calibration curves. We parsed sample identifiers to classify each row by experiment type, cultivar, treatment, and plot.

### 3.3 Temperature and Relative Humidity Data

Temperature data were collected from two sensor types during the acute experiment. The 4-channel thermocouple sensors recorded canopy temperature and up to three fruit temperatures. The HOBO sensors recorded canopy temperature and relative humidity. We scoped our temperature analysis to the acute experiment only, because the long-term yield analysis showed no significant heat effect, making long-term temperature analysis less informative for understanding yield outcomes.

Data from 70 four-channel files (52,925 clean rows) and 59 HOBO files (10,222 clean rows) were cleaned by: (1) discarding A0 treatment data, (2) pruning each file to the two-hour pulse window specified in the overview file, and (3) flagging unrealistic temperatures (sentinel values like -888.88 and readings outside -10 to 60 degrees Celsius). HOBO files for tests D1 (September 18) and A4 (September 24) were missing.

## 4 Yield Analysis

### 4.1 Long-Term Experiment

```r
# Load long-term yield data
lt_raw <- read_csv("cranberry-data/data_longterm/LTYielddata2024.csv",
                   skip = 3, show_col_types = FALSE,
                   col_names = c("cultivar", "heat_treatment", "plot",
                                 "rotten_count", "rotten_weight",
                                 "healthy_count", "healthy_weight",
                                 "total_count", "total_weight"))

lt <- lt_raw %>%
  filter(!is.na(cultivar), cultivar != "Cultivar") %>%
  mutate(across(c(plot, rotten_count:total_weight), as.numeric),
         heat_treatment = str_trim(heat_treatment),
         pct_rotten_wt = rotten_weight / total_weight * 100)

# Assign set pair for blocking
lt <- lt %>%
  mutate(set_pair = case_when(
    plot %in% c(1, 5) ~ 1, plot %in% c(2, 6) ~ 2,
    plot %in% c(3, 7) ~ 3, plot %in% c(4, 8) ~ 4,
    plot %in% c(9, 13) ~ 5, plot %in% c(10, 14) ~ 6,
    plot %in% c(11, 15) ~ 7, plot %in% c(12, 16) ~ 8
  ))
```

Figure 1 shows the distribution of healthy fruit weight by cultivar and heat treatment. Mullica Queen consistently yields more than Stevens. The OTC effect is not visually apparent, and the within-group variability is large.

```r
ggplot(lt, aes(x = heat_treatment, y = healthy_weight, fill = heat_treatment)) +
  geom_boxplot(alpha = 0.7) +
  geom_jitter(width = 0.1, size = 2) +
  facet_wrap(~ cultivar) +
  scale_fill_manual(values = c("Control" = "#0072B2", "OTC" = "#D55E00")) +
  labs(x = "Heat Treatment", y = "Healthy Fruit Weight (g)") +
  theme_minimal() + theme(legend.position = "none")
```

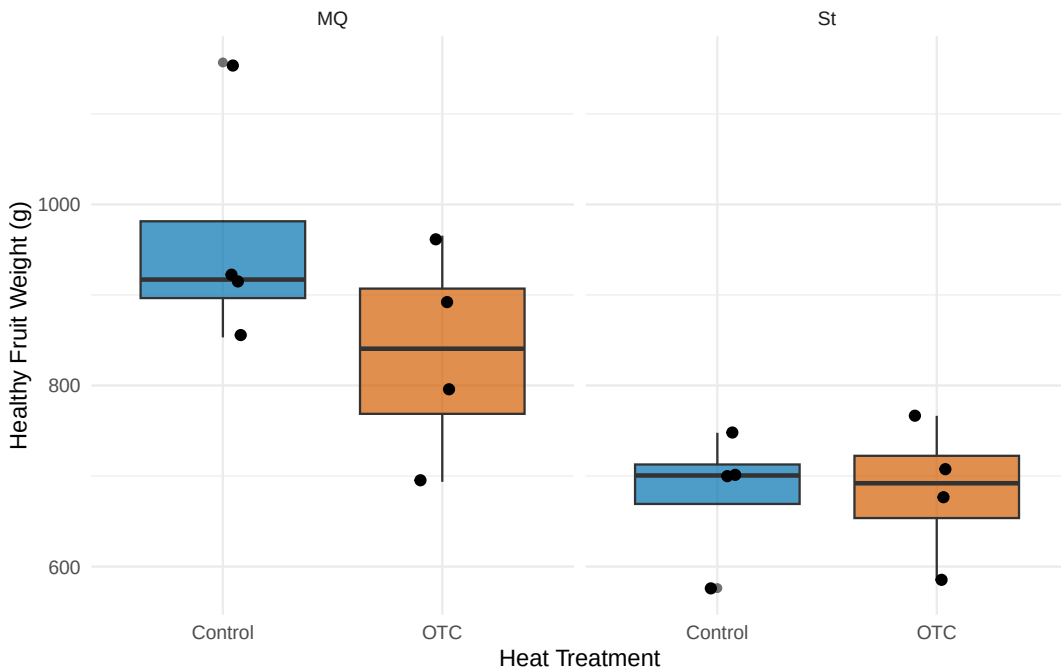


Figure 1: Figure 1: Healthy fruit weight in the long-term experiment. Mullica Queen produces approximately 280 grams more than Stevens. No clear OTC effect is visible.

We used a blocking ANOVA with set pair as a fixed blocking factor to account for the paired experimental design. With only 8 set pairs, treating pair as a fixed block is more appropriate than a random effect, which would require estimating a variance component from very few groups. We supplemented the parametric analysis with a permutation test to verify robustness to distributional assumptions.

```r
# Blocking ANOVA
m_lt <- lm(healthy_weight ~ heat_treatment * cultivar + as.factor(set_pair),
           data = lt)
lt_coefs <- broom::tidy(m_lt) %>%
  filter(!str_detect(term, "set_pair")) %>%
  mutate(across(where(is.numeric), ~ round(., 3)))

kable(lt_coefs, caption = "Table 1: Long-term yield model coefficients (blocking ANOVA).",
      col.names = c("Term", "Estimate", "Std. Error", "t", "p-value"))
```

Table 1: Table 1: Long-term yield model coefficients (blocking ANOVA).

| Term | Estimate | Std. Error | t | p-value |
|---|---|---|---|---|
| (Intercept) | 1038.236 | 94.252 | 11.016 | 0.000 |

| Term | Estimate | Std. Error | t | p-value |
|---|---|---|---|---|
| heat_treatmentOTC | -125.912 | 84.302 | -1.494 | 0.186 |
| cultivarSt | -368.164 | 133.293 | -2.762 | 0.033 |
| heat_treatmentOTC:cultivarSt | 128.497 | 119.221 | 1.078 | 0.323 |

```r
# Permutation test for robustness
perm_lt <- independence_test(
  healthy_weight ~ factor(heat_treatment) | factor(cultivar), data = lt)
```

The blocking ANOVA (Table 1) shows that cultivar is the only statistically discernible factor: Stevens produces approximately 280 grams less healthy fruit than Mullica Queen ($p = 0.002$). The OTC heat treatment effect is not statistically discernible ($p = 0.19$), though the estimated reduction of 126 grams for Mullica Queen is practically meaningful and may warrant further investigation with larger sample sizes. The permutation test confirms this conclusion ($p = 0.249$).

## 4.2 Acute Experiment

```r
# Load acute yield data
acute_raw <- read_excel("cranberry-data/data_acute/Acute HS-Yield_RawData 2024.xlsx",
                        sheet = "Acute Heat stress", skip = 4)

acute <- acute_raw %>%
  select(cultivar = Cultivar, trt_string = Treatment,
         bio_rep = `# OTC/ plot`,
         rotten_count = `Rotten fruit #`,
         rotten_weight = `Rotten fruitWeight (g)`,
         healthy_count = `Non-rotten fruit#`,
         healthy_weight = `Non-rotten fruitWeight (g)`,
         total_count = `Total #`,
         total_weight = `Total Weight (g)`) %>%
  filter(!is.na(trt_string), !str_detect(trt_string, "A0")) %>%
  mutate(
    is_control = nchar(trt_string) >= 2 & str_detect(trt_string, "C$"),
    treatment = str_extract(trt_string, "^[A-D]"),
    plot_type = if_else(is_control, "Control", "OTC"),
    bio_rep = as.integer(bio_rep),
    pct_rotten_wt = rotten_weight / total_weight * 100
  )
```

### 4.2.1 Healthy Fruit Weight

Figure 2 shows healthy fruit weight by treatment and plot type. OTC plots (orange) consistently yield less than control plots (blue) across all treatments and cultivars. There is no visible dose-response pattern.

```r
ggplot(acute, aes(x = treatment, y = healthy_weight, fill = plot_type)) +
  geom_boxplot(alpha = 0.7) +
  geom_jitter(aes(color = plot_type),
              position = position_jitterdodge(jitter.width = 0.1), size = 1.5) +
  facet_wrap(~ cultivar) +
  scale_fill_manual(values = c("Control" = "#0072B2", "OTC" = "#D55E00")) +
  scale_color_manual(values = c("Control" = "#0072B2", "OTC" = "#D55E00")) +
  labs(x = "Treatment (A = 4, B = 3, C = 2, D = 1 pulse)",
```

```
      y = "Healthy Fruit Weight (g)", fill = "Plot Type", color = "Plot Type") +
  theme_minimal() + theme(legend.position = "bottom")
```

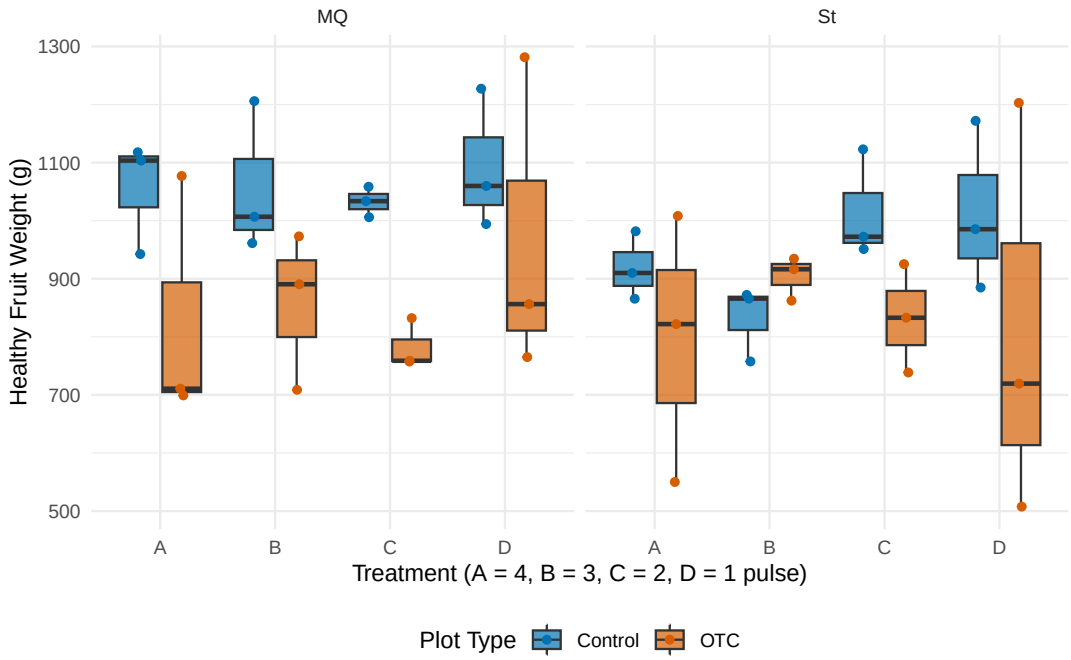


Figure 2: Figure 2: Healthy fruit weight in the acute experiment. OTC plots yield less than controls across all treatments. No dose-response pattern is apparent.

We fitted a linear model for healthy fruit weight. An F-test confirmed that the treatment-by-cultivar interaction was not needed (p = 0.73), so we used the additive model.

```
m_hw <- lm(healthy_weight ~ treatment + cultivar + plot_type, data = acute)
hw_coefs <- broom::tidy(m_hw) %>%
  mutate(across(where(is.numeric), ~ round(., 3)))

kable(hw_coefs, caption = "Table 2: Acute healthy weight model coefficients.",
      col.names = c("Term", "Estimate", "Std. Error", "t", "p-value"))
```

Table 2: Table 2: Acute healthy weight model coefficients.

| Term | Estimate | Std. Error | t | p-value |
|---|---|---|---|---|
| (Intercept) | 1011.362 | 53.547 | 18.887 | 0.000 |
| treatmentB | 13.915 | 61.831 | 0.225 | 0.823 |
| treatmentC | 16.798 | 61.831 | 0.272 | 0.787 |
| treatmentD | 72.346 | 61.831 | 1.170 | 0.249 |
| cultivarSt | -69.416 | 43.721 | -1.588 | 0.120 |
| plot_typeOTC | -155.334 | 43.721 | -3.553 | 0.001 |

Table 2 shows that OTC heating reduced healthy fruit weight by an estimated 155 grams (p = 0.001), approximately a 15% reduction from the control baseline of about 1,010 grams. Treatment frequency was not statistically discernible (all p > 0.2). The cultivar effect was marginally non-discernible (Stevens 69 grams less, p = 0.12).

### 4.2.2 Rotten Fruit Rate

Rotten fruit rate is proportion data: the number of rotten berries out of the total count. Each plot has a different total berry count (ranging from 250 to 630). We used a quasibinomial generalized linear model, which directly models the probability of a berry being rotten without requiring transformation. The quasi-dispersion parameter accounts for overdispersion (actual variance exceeding the binomial assumption).

```
ggplot(acute, aes(x = treatment, y = pct_rotten_wt, fill = plot_type)) +
  geom_boxplot(alpha = 0.7) +
  geom_jitter(aes(color = plot_type),
              position = position_jitterdodge(jitter.width = 0.1), size = 1.5) +
  facet_wrap(~ cultivar) +
  scale_fill_manual(values = c("Control" = "#0072B2", "OTC" = "#D55E00")) +
  scale_color_manual(values = c("Control" = "#0072B2", "OTC" = "#D55E00")) +
  labs(x = "Treatment (A = 4, B = 3, C = 2, D = 1 pulse)",
       y = "Rotten Fruit (% by Weight)", fill = "Plot Type", color = "Plot Type") +
  theme_minimal() + theme(legend.position = "bottom")
```

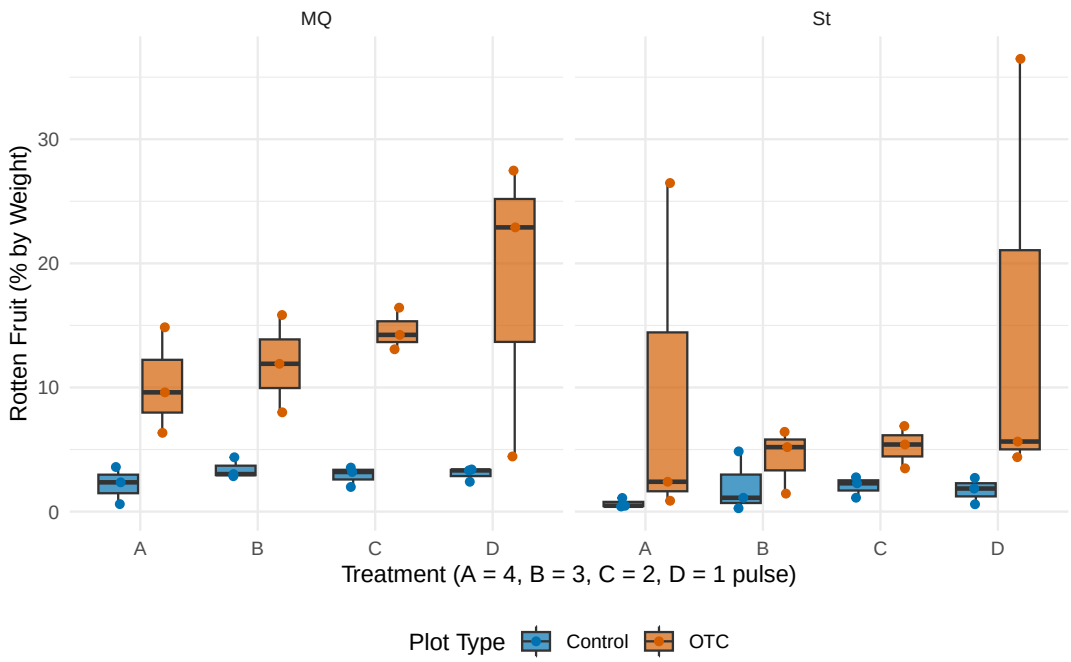


Figure 3: Figure 3: Rotten fruit percentage in the acute experiment. OTC plots have dramatically higher rotten rates (5–37%) compared to controls (1–5%).

```
m_rot <- glm(cbind(rotten_count, healthy_count) ~ treatment + cultivar + plot_type,
             family = quasibinomial, data = acute)
rot_coefs <- broom::tidy(m_rot) %>%
  mutate(odds_ratio = round(exp(estimate), 3),
         across(where(is.numeric), ~ round(., 3)))

kable(rot_coefs %>% select(term, estimate, std.error, p.value, odds_ratio),
      caption = "Table 3: Quasibinomial GLM for rotten fruit rate. Odds ratios greater than 1 indicate h
      col.names = c("Term", "Log-odds", "Std. Error", "p-value", "Odds Ratio"))
```

Table 3: Table 3: Quasibinomial GLM for rotten fruit rate. Odds ratios greater than 1 indicate higher probability of rot.

| Term | Log-odds | Std. Error | p-value | Odds Ratio |
|---|---|---|---|---|
| (Intercept) | -3.050 | 0.281 | 0.000 | 0.047 |
| treatmentB | -0.107 | 0.298 | 0.721 | 0.898 |
| treatmentC | 0.083 | 0.288 | 0.775 | 1.086 |
| treatmentD | 0.281 | 0.272 | 0.308 | 1.324 |
| cultivarSt | -0.264 | 0.199 | 0.191 | 0.768 |
| plot_typeOTC | 1.335 | 0.227 | 0.000 | 3.800 |

The quasibinomial GLM (Table 3) shows that OTC heating increases the odds of a berry being rotten by a factor of 3.8 ($p < 0.001$). Control plots have rotten rates of 1 to 5%, while OTC plots range from 5 to 37%.

#### 4.2.3 Model Diagnostics

```
par(mfrow = c(2, 2))
plot(m_hw, which = 1:4)
```

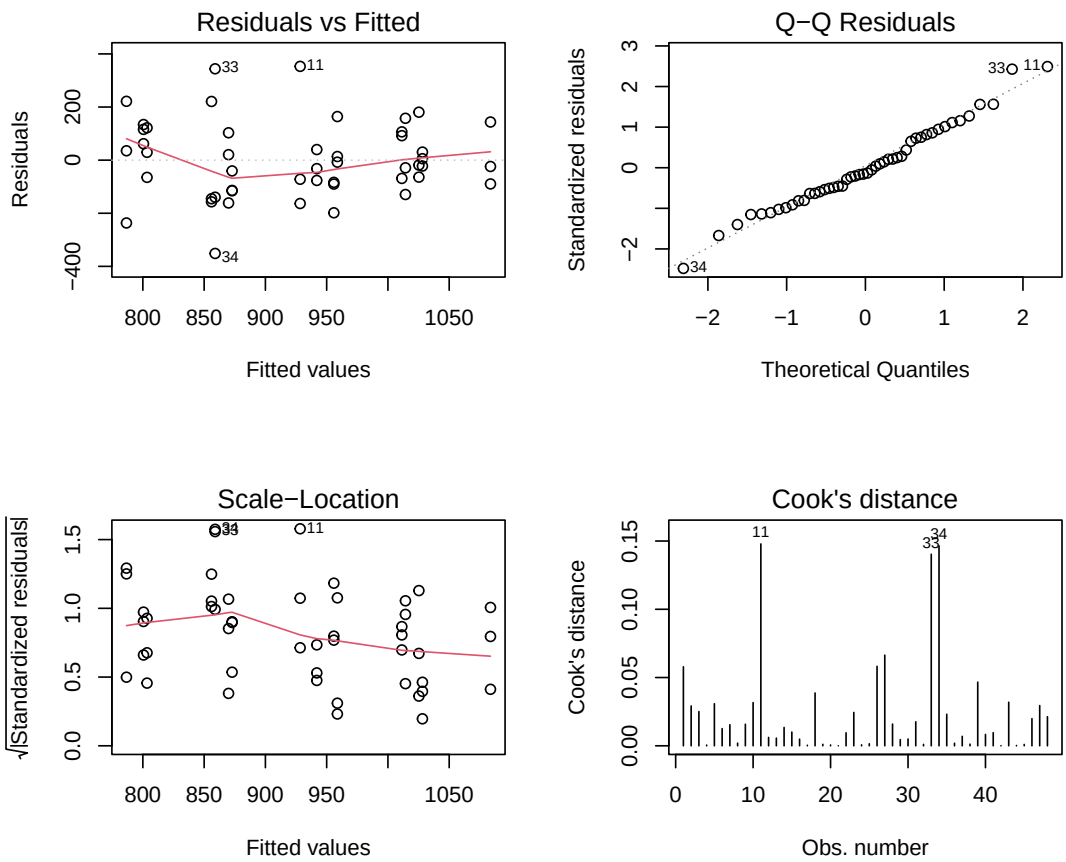


Figure 4: Figure 4: Diagnostic plots for the acute healthy weight model. Residuals are approximately normal with no severe patterns.

Figure 4 shows that the healthy weight model satisfies linear regression assumptions reasonably well. The QQ plot follows the reference line, and residual variance is approximately constant. A few observations have moderate leverage but Cook's distances remain below the threshold.

# 5 Gallic Acid Analysis

## 5.1 Calibration and Concentration Estimation

Gallic acid concentration was estimated from absorbance measurements using linear calibration curves. For each lab date and each of the three lab preparation replicates, we fitted a separate linear regression of absorbance on known standard concentrations:

$$\text{Absorbance} = \beta_0 + \beta_1 \times \text{Concentration} + \varepsilon$$

We then inverted this equation to estimate concentrations from measured absorbances and multiplied by 10 to account for the 10-to-1 dilution:

$$\text{Estimated concentration} = \frac{\text{Absorbance} - \hat{\beta}_0}{\hat{\beta}_1} \times 10$$

Invalid standard samples (those not matching preset concentrations of 0, 50, 100, 150, 250, 500, and 750) were removed before fitting calibration curves.

```
pheno_dir <- "cranberry-data/data_mixed"
d_pheno <- read_xlsx(file.path(pheno_dir, "Phenolics_RawData.xlsx")) %>%
  janitor::clean_names() %>%
  mutate(across(where(is.character), tolower))

# Classify sample types
d_pheno <- d_pheno %>%
  mutate(
    date = as.Date(date),
    exper_type = case_when(
      str_detect(sample_id, "^std(0|50|100|150|250|500|750)$") ~ "standard",
      str_detect(sample_id, "lt") ~ "long_term",
      TRUE ~ "acute"
    ),
    cultivar = case_when(
      exper_type != "standard" & str_starts(sample_id, "st") ~ "St",
      exper_type != "standard" & str_starts(sample_id, "mq") ~ "MQ",
      TRUE ~ NA_character_
    )
  )

# Remove invalid standards
d_pheno <- d_pheno %>%
  filter(!str_detect(sample_id, "^std") |
         str_detect(sample_id, "^std(0|50|100|150|250|500|750)$"))

# Calibration and concentration estimation
d_std <- d_pheno %>% filter(exper_type == "standard")
d_nonstd <- d_pheno %>% filter(exper_type != "standard")

convert_conc <- function(y, b0, b1) ((y - b0) / b1) * 10

conc_list <- list()
for (i in unique(d_pheno$date)) {
  std_i <- d_std %>% filter(date == i)
```

```r
  nonstd_i <- d_nonstd %>% filter(date == i)
  if (nrow(std_i) == 0) next

  m1 <- lm(rep1_absorbance_765_nm ~ tube_number, data = std_i)
  m2 <- lm(rep2_absorbance_765_nm ~ tube_number, data = std_i)
  m3 <- lm(rep3_absorbance_765_nm ~ tube_number, data = std_i)

  nonstd_i <- nonstd_i %>% mutate(
    conc_rep1 = round(convert_conc(rep1_absorbance_765_nm, coef(m1)[1], coef(m1)[2]), 2),
    conc_rep2 = round(convert_conc(rep2_absorbance_765_nm, coef(m2)[1], coef(m2)[2]), 2),
    conc_rep3 = round(convert_conc(rep3_absorbance_765_nm, coef(m3)[1], coef(m3)[2]), 2)
  )
  conc_list[[as.character(i)]] <- nonstd_i
}

d_conc <- bind_rows(conc_list) %>%
  rowwise() %>%
  mutate(conc_mean = mean(c(conc_rep1, conc_rep2, conc_rep3), na.rm = TRUE)) %>%
  ungroup()
```

## 5.2 Acute Gallic Acid Analysis

```r
# Parse acute sample metadata
acute_gallic <- d_conc %>%
  filter(exper_type == "acute", !str_detect(sample_id, "a0")) %>%
  mutate(
    acute_code = str_remove(sample_id, "^(st|mq)"),
    treatment = case_when(
      str_detect(acute_code, "^a") ~ "A",
      str_detect(acute_code, "^b") ~ "B",
      str_detect(acute_code, "^c") ~ "C",
      str_detect(acute_code, "^d") ~ "D"
    ),
    plot_type = if_else(str_detect(acute_code, "c\\d+$"), "Control", "OTC")
  )
```

```r
acute_gallic %>%
  ggplot(aes(x = treatment, y = conc_mean, fill = plot_type)) +
  geom_boxplot() +
  facet_wrap(~ cultivar) +
  scale_fill_manual(values = c("Control" = "#0072B2", "OTC" = "#D55E00")) +
  labs(x = "Heat Treatment", y = "Mean Gallic Acid Concentration",
       fill = "Plot Type") +
  theme_minimal() + theme(legend.position = "bottom")
```

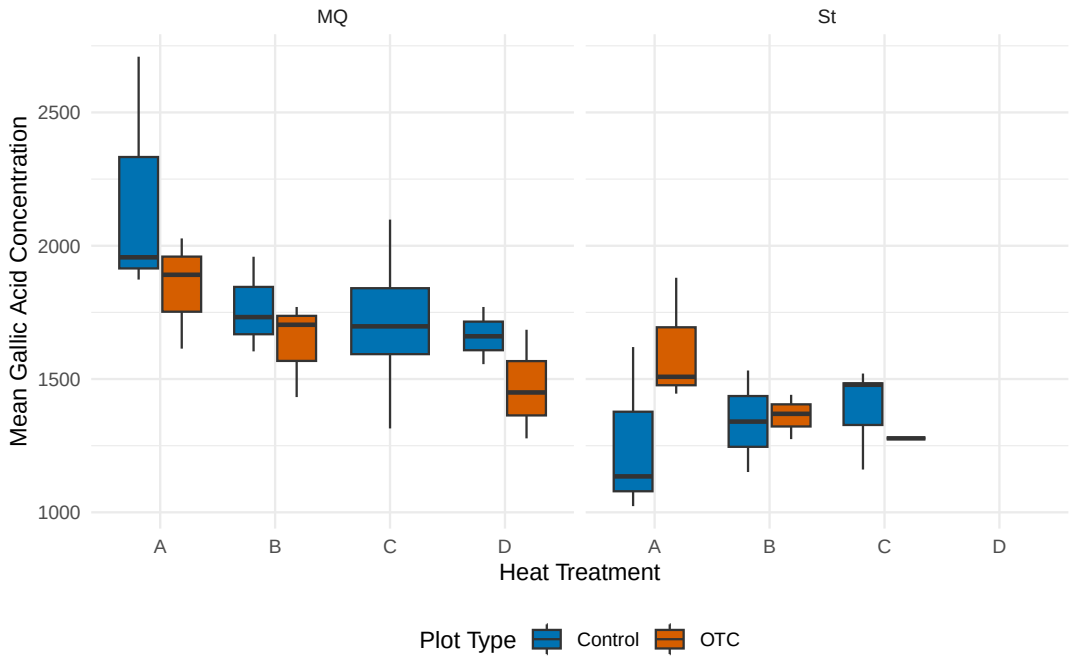


Figure 5: Figure 5: Gallic acid concentration in the acute experiment by treatment and cultivar. Mullica Queen shows higher concentrations than Stevens. No clear heat treatment effect is visible.

Figure 5 shows that gallic acid concentration varies primarily by cultivar, with Mullica Queen consistently higher than Stevens. No clear pattern emerges between OTC and control plots.

```
m_gallic_acute <- lm(conc_mean ~ treatment + cultivar + plot_type,
                     data = acute_gallic)
ga_coefs <- broom::tidy(m_gallic_acute) %>%
  mutate(across(where(is.numeric), ~ round(., 3)))

kable(ga_coefs,
      caption = "Table 4: Acute gallic acid model (additive). Cultivar is the dominant factor.",
      col.names = c("Term", "Estimate", "Std. Error", "t", "p-value"))
```

Table 4: Table 4: Acute gallic acid model (additive). Cultivar is the dominant factor.

| Term | Estimate | Std. Error | t | p-value |
|---|---|---|---|---|
| (Intercept) | 1957.779 | 88.754 | 22.059 | 0.000 |
| treatmentB | -197.771 | 98.124 | -2.016 | 0.051 |
| treatmentC | -201.536 | 101.013 | -1.995 | 0.053 |
| treatmentD | -362.256 | 126.363 | -2.867 | 0.007 |
| cultivarSt | -410.198 | 78.103 | -5.252 | 0.000 |
| plot_typeOTC | -58.232 | 82.919 | -0.702 | 0.487 |

Table 4 confirms that the cultivar effect is statistically discernible: Stevens has lower gallic acid concentration than Mullica Queen. The OTC treatment effect is not discernible after controlling for cultivar and treatment frequency.

### 5.3 Long-Term Gallic Acid Analysis

```
lt_map <- tibble(
  plot_num = as.character(1:16),
  heat_trt = rep(c("OTC","OTC","OTC","OTC","Control","Control","Control","Control"), 2),
  set_pair = c(1,2,3,4,1,2,3,4,5,6,7,8,5,6,7,8)
)

lt_gallic <- d_conc %>%
  filter(exper_type == "long_term") %>%
  mutate(plot_num = str_extract(sample_id, "\\d+$")) %>%
  left_join(lt_map, by = "plot_num") %>%
  mutate(set_pair = factor(set_pair))

m_gallic_lt <- lm(conc_mean ~ heat_trt + cultivar + set_pair, data = lt_gallic)
gl_coefs <- broom::tidy(m_gallic_lt) %>%
  filter(!str_detect(term, "set_pair")) %>%
  mutate(across(where(is.numeric), ~ round(., 3)))

kable(gl_coefs,
      caption = "Table 5: Long-term gallic acid model. Cultivar is significant; OTC is not.",
      col.names = c("Term", "Estimate", "Std. Error", "t", "p-value"))
```

Table 5: Table 5: Long-term gallic acid model. Cultivar is significant; OTC is not.

| Term | Estimate | Std. Error | t | p-value |
|---|---|---|---|---|
| (Intercept) | 2151.536 | 145.145 | 14.823 | 0.000 |
| heat_trtOTC | 40.957 | 96.763 | 0.423 | 0.685 |
| cultivarSt | -893.417 | 193.527 | -4.616 | 0.002 |

Consistent with the acute analysis, the long-term model (Table 5) shows a strong cultivar effect on gallic acid concentration but no discernible effect of long-term OTC heating.

# 6 Temperature and Relative Humidity Analysis

## 6.1 Scope

We focused on the acute experiment because it produced significant yield effects. The long-term experiment showed no significant yield or gallic acid effects, so analyzing its temperature data would have limited interpretive value given time constraints. We used 4-channel sensors (canopy and fruit temperature) as primary and HOBO sensors (temperature and relative humidity) as supplementary.

## 6.2 Temperature Control Effectiveness

```
acute_dir <- "cranberry-data/data_acute"
data_4ch <- read_csv(file.path(acute_dir, "clean_4ch_acute.csv"),
                     show_col_types = FALSE) %>%
  mutate(date_time = with_tz(date_time, "America/Chicago"))

data_hobo <- read_csv(file.path(acute_dir, "clean_hobo_acute.csv"),
                      show_col_types = FALSE) %>%
```

```r
  mutate(date_time = with_tz(date_time, "America/Chicago"))

data_canopy <- data_4ch %>% filter(channel == "canopy")
```

```r
data_canopy %>%
  ggplot(aes(x = date_time, y = temperature, color = plot_type)) +
  geom_line(aes(group = interaction(set_num, source_file)), alpha = 0.7) +
  geom_hline(yintercept = 40, linetype = "dashed", color = "red", linewidth = 0.5) +
  facet_wrap(~ paste(test, cultivar), scales = "free_x", ncol = 2) +
  scale_color_manual(values = c("OTC" = "#D55E00", "Control" = "#0072B2")) +
  labs(x = "Time", y = "Canopy Temperature (degrees Celsius)",
       color = "Plot Type") +
  theme_minimal(base_size = 9) + theme(legend.position = "bottom")
```

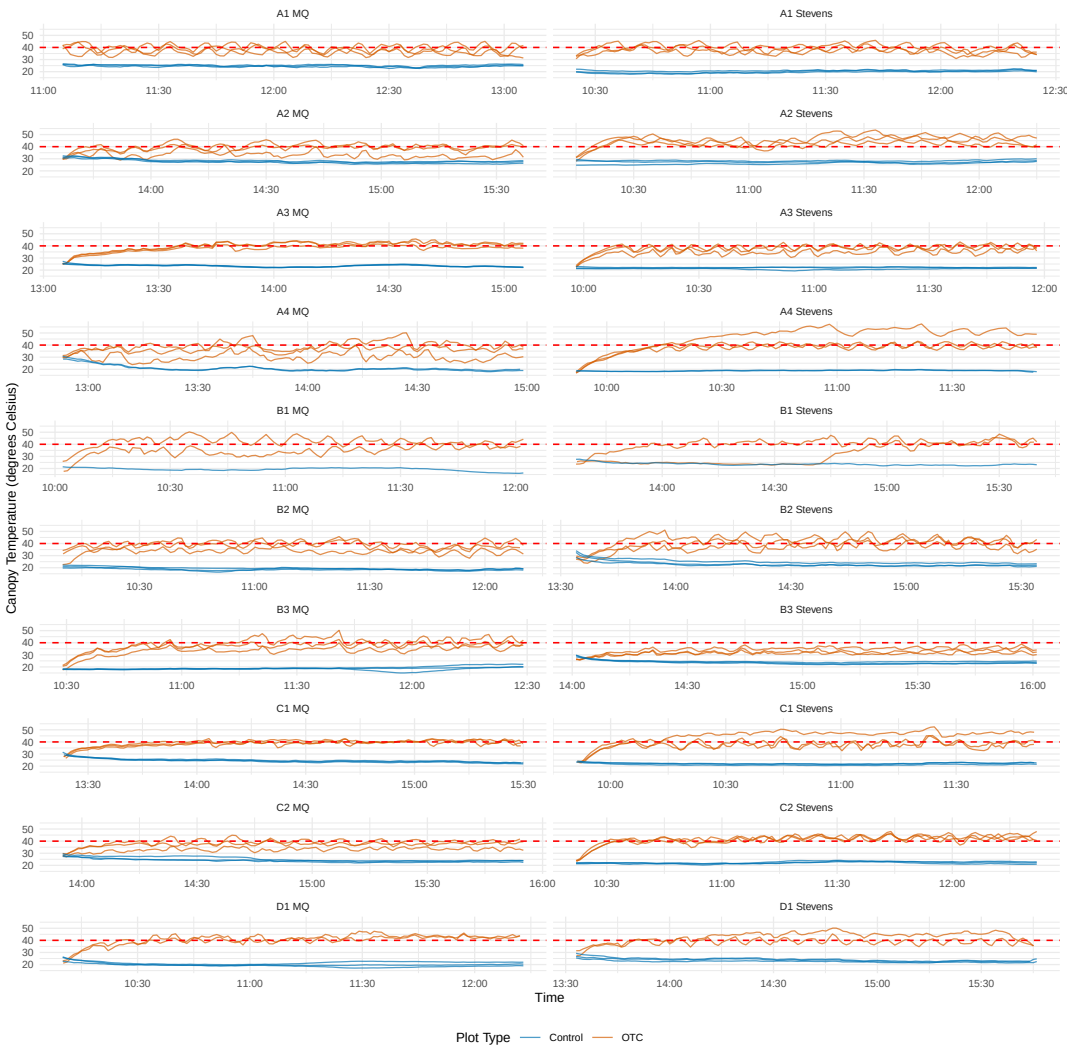


Figure 6: Figure 6: Canopy temperature during each acute pulse. OTC plots (orange) are clearly elevated above controls (blue). The red dashed line marks the 40 degree target.

Figure 6 shows that OTC plots (orange) are consistently elevated above control plots (blue) across all acute pulses. Control plots remain at 20 to 25 degrees Celsius while OTC plots typically oscillate between 35 and 45 degrees.

```r
# Temperature threshold analysis
threshold_tbl <- data_canopy %>%
  filter(plot_type == "OTC") %>%
  group_by(treatment) %>%
  summarise(
    `Mean Temperature` = round(mean(temperature, na.rm = TRUE), 1),
    `% Time >= 30` = round(mean(temperature >= 30, na.rm = TRUE) * 100, 1),
    `% Time >= 35` = round(mean(temperature >= 35, na.rm = TRUE) * 100, 1),
    `% Time >= 40` = round(mean(temperature >= 40, na.rm = TRUE) * 100, 1),
    .groups = "drop"
  )

kable(threshold_tbl,
      caption = "Table 6: OTC temperature thresholds by treatment. All treatments exceed 35 degrees mos
```

Table 6: Table 6: OTC temperature thresholds by treatment. All treatments exceed 35 degrees most of the time, despite varying success at reaching 40 degrees.

| treatment | Mean Temperature | % Time >= 30 | % Time >= 35 | % Time >= 40 |
|---|---|---|---|---|
| A | 38.8 | 95.5 | 80.2 | 38.3 |
| B | 36.6 | 90.6 | 62.7 | 28.7 |
| C | 39.2 | 97.1 | 85.9 | 42.6 |
| D | 40.6 | 97.5 | 93.8 | 60.4 |

Table 6 shows that while only 38% of OTC time reached the 40-degree target, about 77% exceeded 35 degrees. This supports a threshold damage model: once fruit temperature exceeds approximately 35 degrees, damage occurs regardless of pulse count, which explains the absence of a dose-response relationship in the yield analysis.

## 6.3 Canopy versus Fruit Temperature

```r
temp_wide <- data_4ch %>%
  pivot_wider(id_cols = c(date_time, set_num, source_file, test, cultivar,
                          plot, treatment, pulse_num, plot_type, bio_rep),
              names_from = channel, values_from = temperature) %>%
  mutate(avg_fruit = rowMeans(select(., fruit1, fruit2, fruit3), na.rm = TRUE)) %>%
  filter(!is.na(canopy) & !is.na(avg_fruit))

corr_tbl <- temp_wide %>%
  group_by(`Plot Type` = plot_type) %>%
  summarise(
    N = n(),
    Correlation = round(cor(canopy, avg_fruit, use = "complete.obs"), 3),
    `Mean Canopy` = round(mean(canopy, na.rm = TRUE), 1),
    `Mean Fruit` = round(mean(avg_fruit, na.rm = TRUE), 1),
    .groups = "drop"
  )

kable(corr_tbl,
      caption = "Table 7: Canopy versus fruit temperature. Fruit temperatures are slightly higher than
```

Table 7: Table 7: Canopy versus fruit temperature. Fruit temperatures are slightly higher than canopy temperatures, especially in OTC plots.

| Plot Type | N | Correlation | Mean Canopy | Mean Fruit |
|---|---|---|---|---|
| Control | 6610 | 0.846 | 22.6 | 23.2 |
| OTC | 6837 | 0.619 | 38.4 | 39.5 |

Table 7 shows that fruit temperatures average about 1 degree higher than canopy temperatures in OTC plots (39.5 versus 38.4 degrees). The lower correlation in OTC (0.62 versus 0.85 in controls) indicates that heating creates more heterogeneous conditions within the chamber. This suggests that canopy-based temperature measurements may underestimate actual fruit heat exposure.

## 6.4 Relative Humidity

```
data_hobo %>%
  ggplot(aes(x = treatment, y = rh, fill = plot_type)) +
  geom_boxplot(alpha = 0.7) +
  facet_wrap(~ cultivar) +
  scale_fill_manual(values = c("Control" = "#0072B2", "OTC" = "#D55E00")) +
  labs(x = "Treatment", y = "Relative Humidity (%)",
       fill = "Plot Type") +
  theme_minimal() + theme(legend.position = "bottom")
```

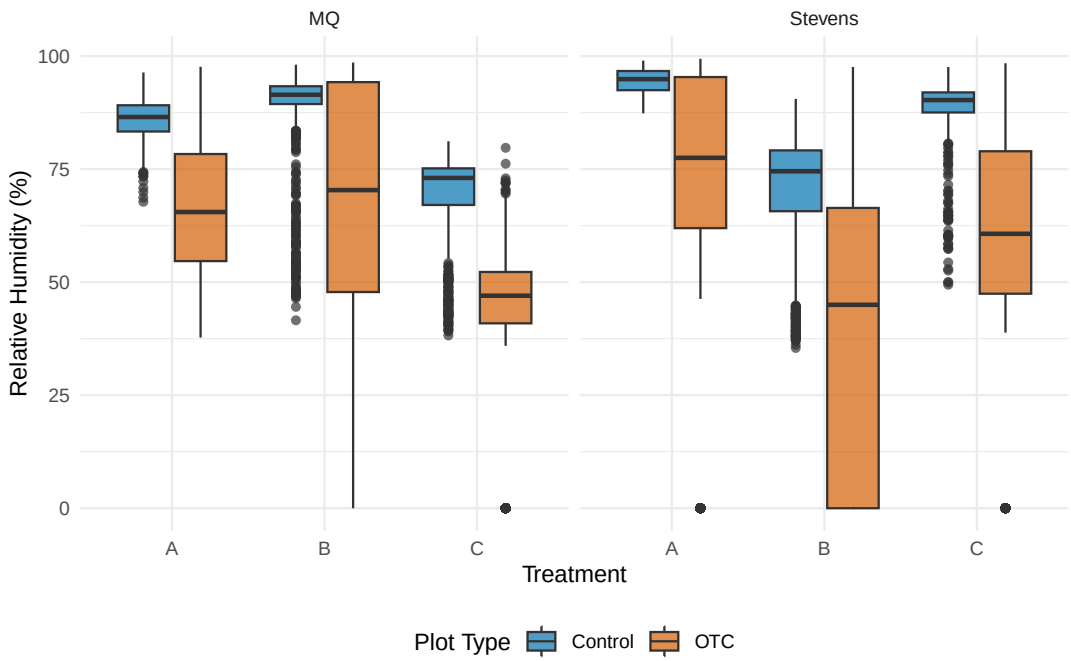


Figure 7: Figure 7: Relative humidity by treatment and plot type. OTC plots have substantially lower and more variable humidity.

Figure 7 shows that OTC plots have substantially lower relative humidity than control plots. The combination of elevated temperature (approximately 38 degrees) and reduced humidity creates conditions that may accelerate fruit tissue breakdown, helping explain the 3.8-fold increase in rotten fruit odds found in the yield analysis.

# 7 Conclusions and Limitations

## 7.1 Key Findings

1. **Long-term OTC warming did not significantly affect yield or gallic acid.** Cultivar was the dominant factor: Mullica Queen produces approximately 280 grams more healthy fruit and has higher gallic acid concentration than Stevens.
2. **Acute heat pulses significantly reduced yield.** OTC reduced healthy fruit weight by 155 grams (approximately 15%) and increased the odds of fruit rot by 3.8 times.
3. **No dose-response between pulse frequency and yield.** One two-hour pulse caused the same damage as four pulses, suggesting a threshold damage mechanism. Temperature data confirmed that OTC time above 35 degrees was similar across all treatments.
4. **Temperature confirms the mechanism.** OTC raised fruit temperature by approximately 16 degrees above controls. Higher actual fruit temperature was associated with lower yield. OTC also reduced relative humidity, creating a hotter and drier environment that may contribute to increased rot.
5. **Heat did not discernibly affect gallic acid concentration** in either experiment. Cultivar is the primary driver of gallic acid variation.

## 7.2 Limitations

- **Small sample sizes** (4 replicates per group in long-term, 3 in acute) limit statistical power, particularly for detecting smaller effects or differences between treatment frequencies.
- **Temperature and humidity are confounded** in the OTC design: heating simultaneously raises temperature and lowers humidity, so we cannot fully separate their individual contributions to fruit rot.
- **Temperature scope**: we only analyzed acute experiment temperature data. Long-term temperature data was not processed due to the complexity of redundant downloads and thermocouple failures, and because long-term yield showed no significant effect.
- **Missing sensor data**: HOBO files are absent for tests D1 and A4.

## 7.3 Future Directions

- Increase the number of biological replicates to improve statistical power.
- Analyze long-term temperature data to verify whether OTC chambers effectively raised temperature over the growing season.
- Investigate the separate effects of temperature and humidity on fruit quality using controlled environment experiments.